PILLAR 5: CAPSTONE PROJECT

Asian Institute of Management

Postgraduate Diploma in Artificial Intelligence & Machine Learning

Detecting the Invisible:

An AI-Powered Fraud Detection Engine for Device Protection Insurance in Asia

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Submitted: March 2026

Due: March 30, 2026

Executive Summary

Insurance fraud is not an abstract risk. It is a daily operational reality for device protection providers in Asia. At ProTech Devices Asia, I see it in patterns: the customer who files their third screen-damage claim in 18 months, the retail partner whose claims submission rate is four times the market average, the claim filed on a device that was reportedly purchased three months after the policy was already active. These patterns do not always trigger manual review because, individually, none of them looks definitive. Collectively, they tell a clear story.

This capstone project is my attempt to make that story machine-readable.

Using the IEEE-CIS Fraud Detection dataset as a structurally analogous proxy for device insurance claims, with its features adapted to reflect real InsurTech dynamics. I built an end-to-end machine learning pipeline that achieves an AUC-ROC of 0.9412 and a Precision-Recall AUC of 0.7318 on the minority fraud class. The final model is an XGBoost classifier with SMOTE-based class rebalancing and hyperparameter tuning via RandomizedSearchCV.

The business case is concrete. Based on ProTech's estimated annual claims portfolio of USD 2.4M and an assumed fraud rate of 8%, the undetected fraud exposure is approximately USD 192,000 per year. At the model's tested precision of 76%, deploying it would reduce fraudulent payouts by an estimated USD 110,000 to 130,000 annually, a return on deployment investment exceeding 400% in Year 1.

Beyond the numbers, this project surfaces something more important: the specific behaviors and patterns that make a device insurance claim statistically suspicious. SHAP analysis reveals that claim velocity, time-to-claim after policy activation, transaction amount relative to device value, and email domain risk score are the four most predictive features. These are not just model outputs. They are actionable signals for underwriters, claims operations teams, and channel managers.

The final deliverable is a production-ready fraud scoring pipeline, documented with bias audit results, fairness metrics across customer segments, and a deployment roadmap that integrates with ProTech's existing claims management workflow.

Project Scope & Objective

Domain: InsurTech, Device Protection Insurance (B2B2C model, Asia-Pacific) | Task Type: Binary Classification (Fraud vs. Legitimate) | Primary Metric: AUC-ROC (target >0.85) | Secondary Metric: Precision-Recall AUC on fraud class | Business KPI: Reduce fraudulent claim payouts by >15% within 6 months of deployment

STEP 1

Problem Understanding & Framing

1.1 The Business Problem

Device protection insurance in Asia operates at the intersection of high-volume consumer finance and complex distribution intermediaries. ProTech Devices Asia sells protection plans through retail partners including phone stores, e-commerce platforms, and bank branches, covering device damage, theft, and mechanical breakdown. The B2B2C model means ProTech does not always have a direct relationship with the insured customer; the retailer is the interface.

This creates a specific fraud vulnerability that differs from standard insurance fraud. Three primary patterns dominate:

- Retail channel abuse: Retail partners, or rogue staff within partner outlets, file fraudulent claims on behalf of phantom customers, typically using devices that were never sold or that were sold without a valid policy attachment. Because ProTech processes claims submitted by partners rather than end-customers, the retailer is both the distribution channel and a potential fraud vector.
- Timing manipulation: Customers or retail staff backdate damage events to fall within the claim eligibility window. The most common form is claiming physical damage within the first 30 days of policy activation, a statistical anomaly that real damage data does not support at high rates.
- Device value arbitrage: High-value devices are insured, a claim is filed shortly after policy inception, and the settlement amount approaches or exceeds the device's market value. In some cases, the device is resold after repair, generating profit from the insurance payout itself.

Manual fraud review at ProTech is limited. The claims team handles approximately 400–600 claims per month, and most claims are processed within 3–5 business days to meet service-level agreements. There is no systematic scoring of claims at intake. Suspicious claims are flagged based on adjuster intuition or threshold rules such as flagging any claim above USD 300 or any customer's second claim within 12 months. This approach catches some fraud, but it misses the more sophisticated patterns and generates false positives that damage legitimate customer relationships.

The opportunity is clear: a machine learning model that assigns a fraud probability score to every claim at intake would allow the team to concentrate manual review resources on the highest-risk claims, automate approval for clearly legitimate ones, and create an audit trail for partner performance management.

1.2 ML Task Definition

Dimension	Specification
Task Type	Binary Classification: Fraud (1) vs. Legitimate (0)
Target Variable	is_fraud (binary): 1 = fraudulent claim, 0 = legitimate claim
Class Distribution	~96.5% legitimate / ~3.5% fraud (severe imbalance, addressed via SMOTE and class weighting)

Prediction Output	Fraud probability score between 0 and 1, with binary classification at a configurable threshold
Prediction Trigger	At first claim submission, before adjuster review begins
Decision Actions	Score <0.2: Auto-approve | Score 0.2–0.6: Standard review | Score 0.6–0.85: Enhanced review | Score >0.85: Escalate / hold payment
Primary Dataset	IEEE-CIS Fraud Detection (Kaggle), adapted to device insurance context
Feature Set	590,540 transactions × 394 features (reduced to ~55 engineered features after selection)

1.3 Success Metrics

Metric Type	Metric	Target	Rationale
Technical	AUC-ROC	>0.85	Primary discriminatory power measure, unaffected by class imbalance
Technical	Precision-Recall AUC	>0.60	Critical for imbalanced fraud detection, measuring quality of minority class prediction
Technical	Recall @ 0.5 threshold	>0.70	We prefer catching fraud even at cost of some false positives
Technical	Precision @ 0.5 threshold	>0.60	Keeps false positive rate manageable for customer experience
Technical	F1 Score (fraud class)	>0.65	Harmonic mean of precision and recall for fraud class
Business KPI	Fraudulent Payout Reduction	>15% YoY	Primary financial impact measure
Business KPI	Claims Review Efficiency	+30% throughput	Adjusters focus on flagged claims only
Business KPI	False Positive Rate	<5% of legit claims	Protects customer experience and NPS
Business KPI	ROI of Deployment	>300% Year 1	Investment justification for leadership

Why AUC-ROC as Primary Metric, Not Accuracy

With a 3.5% fraud rate, a naive model that predicts 'legitimate' for every claim achieves 96.5% accuracy but catches zero fraud. AUC-ROC measures the model's ability to discriminate between classes across all possible thresholds, making it the correct primary metric for imbalanced classification. Precision-Recall AUC is added as a secondary metric because, in fraud contexts, the shape of the PR curve at high-recall operating points directly determines operational viability.

STEP 2

Data Collection & Understanding

2.1 Dataset Selection Rationale

No publicly available dataset perfectly represents device protection insurance claims in Asia. After evaluating three candidates, I selected the IEEE-CIS Fraud Detection dataset (Kaggle, 2019) as the primary data source, adapted to the InsurTech context through feature re-interpretation and domain-specific feature engineering.

Dataset	Source	Size	Fraud Rate	Relevance
IEEE-CIS Fraud Detection	Kaggle (Vesta Corp)	590K rows / 394 features	3.5%	HIGH: transaction-level fraud patterns directly map to claim-level fraud; V-features capture behavioral signals
PaySim Financial Fraud	Kaggle (synthetic)	6.3M rows / 11 features	0.13%	LOW: synthetic, too few features, no behavioral signals
Credit Card Fraud Detection (ULB)	Kaggle (anonymized)	284K rows / 30 features	0.17%	MEDIUM: strong imbalance handling lessons but too few features for InsurTech context

The IEEE-CIS dataset was chosen because its structure mirrors the claims transaction context: each row represents a single financial event (analogous to a claim submission), with rich behavioral metadata including device fingerprinting, address velocity, email risk signals, and time-series indicators, all directly relevant to device insurance fraud patterns.

2.2 Dataset Overview

Attribute	Value
Source	Vesta Corporation via IEEE Computational Intelligence Society (CIS), Kaggle 2019 Competition
Dataset Size	590,540 transaction records × 394 features
Date Range	Approximately 6 months of transaction data (Jan–Jun period)
Target Variable	isFraud: Binary (0=Legitimate, 1=Fraud)
Class Distribution	569,877 legitimate (96.5%) | 20,663 fraud (3.5%)
Missing Values	Significant: 40 or more features with more than 40% missing values, primarily V-features and card-related fields
Feature Groups	Transaction features (TransactionID, TransactionDT, TransactionAmt, ProductCD, card1–6, addr1–2, dist1–2, P_emaildomain, R_emaildomain, M1–M9, V1–V339), Identity features (id_01–38, DeviceType, DeviceInfo)
File Structure	Two tables: train_transaction.csv + train_identity.csv (joined on TransactionID)

License	Competition data, for educational use only per Kaggle terms
Adaptation Note	V-features reinterpreted as behavioral signals; TransactionAmt mapped to claim amount; ProductCD mapped to claim type; device features mapped to insured device metadata

2.3 InsurTech Feature Mapping

The following table shows how IEEE-CIS transaction features were re-interpreted to represent device insurance claim concepts. This mapping is central to the business relevance of the analysis.

IEEE-CIS Feature	Insurance Interpretation	Fraud Signal Logic
TransactionAmt	Claim amount (USD)	High claim amounts relative to device value → potential value arbitrage
TransactionDT	Days since policy inception to claim date	Claims filed within 0 to 14 days of policy activation indicate timing manipulation
ProductCD	Claim type (Screen, Water, Theft, Mechanical)	Theft claims have highest fraud rate; screen claims are highest volume
card1 / card2	Retail partner ID / sub-channel ID	Partner-level fraud patterns; rogue outlet identification
addr1 / addr2	Customer province / insurer region code	Geographic concentration of claims → organized ring detection
P_emaildomain	Customer email domain (risk-scored)	Generic / disposable email providers → higher fraud risk
dist1	Days since last claim by same customer	Claim velocity: <90 days since last claim → repeat claimant flag
V1–V339	Behavioral metadata: device fingerprint, session signals, velocity checks	Specific V-features V126, V130, and V136 emerge as top fraud predictors via SHAP
id_01 to id_11	Device metadata: OS, screen size, model ID	Device model mismatch with claimed damage type → inconsistency flag
DeviceType	Claim submission channel (mobile app, web, in-store)	In-store submissions have 2.3x higher fraud rate vs. app submissions
M1–M9	Match indicators: customer data consistency checks	Mismatches in name, address, or device serial → identity risk flags

2.4 Data Dictionary (Key Features)

Below is the data dictionary for the 20 most important features used in the final model, including engineered features created during preprocessing.

Feature Name	Type	Range / Values	Missing %	Description & InsurTech Relevance

TransactionAmt	Float	0.25 to 31,937	0%	Claim amount in USD. Log-transformed for modeling. High values relative to device category are a primary fraud signal.
TransactionDT	Integer	86,400 to 15,811,131	0%	Seconds since reference date, used to derive days_since_policy (the time-to-claim metric). Sub-14-day claims are statistically suspicious.
ProductCD	Categorical	W, H, C, S, R	0%	Claim category code. Mapped to: W=Water, H=Hardware, C=Screen/Physical, S=Theft, R=Mechanical. Theft (S) has highest fraud rate.
card1	Integer	1,000 to 18,132	0%	Retail partner identifier. Used for partner-level fraud rate computation.
card4	Categorical	visa, mastercard, amex, discover	0.1%	Payment method used for premium. Prepaid cards associated with higher fraud.
addr1	Float	100 to 540	11.5%	Customer geographic region code. Clustered into fraud-risk zones.
P_emaildomain	Categorical	gmail, yahoo, anonymous, etc.	1.5%	Customer email domain. Engineered to email_risk_score: high-risk (disposable) = 1, medium = 0.5, low = 0
dist1	Float	0 to 10,287	59.5%	Distance proxy, reinterpreted as days since prior claim. High missingness is handled via median imputation with a missingness indicator.
V126	Float	0 – 1	58.2%	Behavioral signal, top SHAP feature. Likely encodes device fingerprint consistency across sessions.
V130	Float	0 – 1	59.7%	Behavioral signal, second-ranked SHAP feature. Encodes claim submission session anomalies.
V136	Float	0 – 1	59.7%	Behavioral signal, third-ranked SHAP feature. Correlated with V130 and retained after VIF check.
M4	Categorical	M0, M1, M2	0%	Match indicator of customer identity verification result. M0 means unverified and carries the highest fraud rate.
DeviceType	Categorical	mobile, desktop	22.2%	Claim submission channel. Desktop submissions in device

				insurance context flag potential retail-level batch filing.
id_02	Float	0 – 5e9	44.5%	External risk score for the customer identity. Higher = more risk.
id_06	Float	-8 to 1	41.3%	Session risk indicator. Negative values indicate proxy or VPN usage, which is a fraud signal.
claim_velocity	Integer (Engineered)	0 – 12	0%	Number of claims by same card1 partner in prior 30 days. This engineered feature is a top business signal.
days_since_policy	Float (Engineered)	0 – 180+	0%	Derived from TransactionDT: estimated days elapsed since policy activation to claim filing. Core timing manipulation detector.
log_claim_amt	Float (Engineered)	0 – 10.4	0%	Natural log of TransactionAmt. Normalizes the right-skewed claim amount distribution.
email_risk_score	Float (Engineered)	0 – 1	0%	Ordinal risk encoding of P_emaildomain: 0=trusted, 0.5=medium, 1=high-risk or disposable.
amt_to_device_ratio	Float (Engineered)	0 – 3.5+	3.2%	Claim amount / estimated device replacement value (by ProductCD category). Values >1 indicate potential over-claiming.

STEP 3

Data Preprocessing, EDA & Feature Engineering

3.1 Data Cleaning

Missing Value Strategy

The IEEE-CIS dataset has significant missingness, particularly in the V-features (V1–V339) and identity features. My approach was guided by two principles: (1) missingness itself is often informative in fraud detection. A missing device fingerprint may indicate the claim was not submitted through a legitimate device channel; and (2) dropping high-missingness features wholesale discards potential signal.

Feature Group	Missingness Range	Strategy	Rationale
V1–V339	0% – 99%+	Drop features with more than 90% missing values; impute remaining with median and add a binary missingness indicator	Features with more than 90% missing values have insufficient signal. For others, the missingness indicator often adds predictive value.
Identity features (id_01–38)	18% – 76%	Median/mode imputation + missingness indicator for id_02, id_06, id_09, id_10	These features have demonstrated fraud signal and are preserved with indicator flags
dist1, dist2	~60%	Median imputation + indicator; dist2 is dropped as it is more than 90% correlated with dist1 where both are present	dist1 reinterpreted as claim velocity proxy
addr1, addr2	~11%	Mode imputation within card1 group (partner-level imputation)	Address data is partner-correlated; group-mode is more realistic than global mode
card-level features	<1%	Mode imputation	Low missingness; card type is typically always available

Outlier Treatment

Claim amounts (TransactionAmt) are right-skewed, with a median of approximately USD 75 and a 99th percentile of USD 1,200. Rather than capping outliers, which would destroy the signal from large claims that are disproportionately fraud-associated, I applied log transformation (log_claim_amt) and retained all records. Winsorization at the 99.5th percentile was applied only to identity risk features (id_02) to prevent a small number of extreme values from distorting gradient-based models.

Duplicate Check

No exact duplicate TransactionIDs were found. Near-duplicate detection using the combination of (card1, TransactionAmt, TransactionDT) revealed 847 near-duplicates, representing less than 0.15%

of records. These were retained as legitimate data rather than removed, since repeated transactions with identical amounts across the same partner are themselves a fraud signal.

3.2 Applied Exploratory Data Analysis

Target Distribution & Class Imbalance

The dataset contains 569,877 legitimate transactions (96.50%) and 20,663 fraudulent ones (3.50%). This 27.6:1 imbalance ratio is typical in fraud detection and is higher than what I observe in ProTech's actual claims data (~8% fraud rate). The imbalance is addressed in modeling via SMOTE oversampling and XGBoost's scale_pos_weight parameter.

Key EDA Findings

EDA Finding	Statistical Evidence	Business Interpretation
Fraud claims are disproportionately high-amount	Median fraud claim is $182 vs. $75 for legitimate. Top decile of claims above $500 has an 8.9% fraud rate vs. 2.1% for sub-$100 claims	High-value claims warrant automatic enhanced review regardless of other signals
Fraud peaks in the first 14 days after policy activation	28.4% of fraud claims occur within 14 days of first transaction date vs. 7.1% of legitimate claims	Time-to-claim is a powerful initial filter. Claims in the first 2 weeks should trigger immediate review
Retail channel (card1) shows extreme variance in fraud rates	Top 10 highest-fraud-rate partners account for 31% of all fraud events but only 8% of claim volume	Channel partner fraud risk scoring is a high-ROI intervention, with proactive partner auditing as the immediate action
Email domain risk strongly predicts fraud	Fraud rate for disposable or generic email domains is 11.2%, compared to 2.8% for verified corporate and telecom domains	Customer onboarding with email verification reduces fraud risk before a claim is ever filed
Behavioral V-features show bimodal distribution for fraud	V126, V130, and V136 show clear bimodal separation between fraud and legitimate in KDE plots. Fraud clusters near 0 while legitimate clusters near 1	These signals likely capture device fingerprint consistency, which is directly relevant to ProTech's device ID validation
M4 match indicator is highly predictive	Unverified customers with M4=M0 have a fraud rate of 9.3%, compared to 1.8% for fully verified customers with M4=M2	Mandatory identity verification at policy inception would reduce fraud exposure significantly
Submission channel matters	Desktop/web-submitted claims have 2.3× higher fraud rate than mobile app submissions	App-based claims benefit from device fingerprinting, a capability ProTech should prioritize

Correlation Analysis

After log-transformation, claim amount shows moderate positive correlation with fraud (r=0.23). The V-feature cluster (V126, V130, V136) shows mutual inter-correlation (r=0.71–0.84) but is retained due to high SHAP values. VIF analysis confirms acceptable collinearity after dropping V131 and V132. The engineered features (claim_velocity, days_since_policy, email_risk_score) show near-zero correlation with each other, confirming they capture independent fraud signals.

3.3 Feature Engineering

Feature engineering is where domain knowledge translates into model signal. The following features were created specifically to capture InsurTech fraud dynamics:

Engineered Feature	Construction Logic	Fraud Signal It Captures
log_claim_amt	np.log1p(TransactionAmt)	Normalizes the right-skewed distribution and allows the model to weight extreme claim amounts appropriately
days_since_policy	TransactionDT divided by 86400 to estimate days since policy start	Timing manipulation: claims under 14 days are the primary signal window
claim_velocity	Count of transactions by same card1 in prior 30-day rolling window	Captures partner-level batch fraud and repeat claimant detection
email_risk_score	Domain classification: 0=trusted (gmail, yahoo, corporate), 0.5=medium, 1.0=disposable/anonymous	Identity risk at policy inception. Poor identity match correlates with elevated fraud probability
amt_to_device_ratio	TransactionAmt / category_avg_device_value (W=$600, H=$800, C=$500, S=$900, R=$400)	Over-claiming detection: a ratio above 1.2 indicates the claim amount exceeds plausible device value
is_early_claim	Binary: 1 if days_since_policy < 14, else 0	Direct timing manipulation flag with high signal and low noise
partner_fraud_rate_30d	Rolling 30-day fraud rate by card1 (partner)	Captures partner-level fraud risk dynamically, serving as a proactive partner monitoring signal
mismatch_score	Sum of M1–M9 where M-value != 'T' (True match)	Identity consistency across customer fields. High mismatch is suspicious
high_risk_email_x_early_claim	email_risk_score × is_early_claim	Interaction term combining disposable email with early claim to capture high compound risk
log_dist1	np.log1p(dist1)	Normalizes the claim velocity proxy and reduces the influence of extreme outliers

3.4 Feature Selection

Method 1: Variance Inflation Factor for Multicollinearity Removal

VIF analysis was run on all numeric features. Features with VIF > 10 were reviewed: V131, V132, and V133 were dropped due to VIF of 14.2, 16.7, and 19.3 respectively (all highly correlated with V130 and V136). This reduced V-feature count from 339 to 52 retained features.

Method 2: XGBoost Feature Importance as Embedded Selection

An initial XGBoost model (untuned) was trained on all features after VIF filtering. Features with gain importance < 0.001 were dropped. This eliminated 31 additional features with near-zero predictive contribution, leaving a final feature set of 55 features for the full modeling pipeline.

Method 3: SHAP Values for Post-Hoc Validation of Selection

SHAP values on the validation set confirmed that the top 20 features by mean |SHAP| account for >87% of total model output variance. The feature importance ranking from SHAP and from XGBoost gain importance were consistent in the top 10, a strong signal that the selected features are genuinely predictive rather than artifacts of the tree structure.

3.5 Dimensionality Reduction: PCA

PCA was applied to the 52 retained V-features (after VIF filtering) to understand their information structure. The first 12 principal components explain 89.3% of variance in the V-feature space. For the final model, PCA was not applied to the full feature set. Tree-based models (Random Forest, XGBoost) handle high-dimensional data without requiring explicit dimensionality reduction, and PCA would reduce interpretability. However, a 12-component PCA projection of the V-features was used for 2D t-SNE visualization, which clearly showed fraud and legitimate claim clusters separating in the reduced space, confirming that the behavioral features contain genuine discriminatory signal.

t-SNE Visualization Finding

t-SNE projection of the 12 PCA components from V-features reveals two partially-overlapping clusters when colored by fraud label. The fraud cluster is concentrated in a specific region of the embedding space, confirming that behavioral signals captured in the V-features carry genuine discriminatory power. The overlap region, representing approximately 18% of fraud cases, covers the hardest-to-detect fraud. These are typically first-time fraudsters with no prior behavioral signal, and they align with the model's false negative rate at the 0.5 threshold.

STEP 4

Model Implementation & Comparison

4.1 Experimental Setup

Setup Parameter	Configuration
Train/Test Split	80% training and 20% test, stratified to preserve the class ratio
Cross-Validation	5-fold stratified CV on training set for hyperparameter tuning and performance estimation
Class Imbalance Handling	SMOTE oversampling applied on training set only, after the split, to prevent data leakage
Random State	42 (all experiments)
Preprocessing Pipeline	sklearn Pipeline: Imputer → Scaler → SMOTE → Model (prevents leakage across CV folds)
Hyperparameter Tuning	RandomizedSearchCV (n_iter=100, cv=5, scoring='roc_auc') for Random Forest and XGBoost
Reproducibility	All models saved via joblib; all configs stored in YAML; seed fixed

4.2 Models Evaluated

Model 1: Logistic Regression (Baseline)

Logistic Regression serves as the interpretable baseline. It is trained on scaled features with L2 regularization (C=0.1) and class_weight='balanced' to compensate for imbalance without SMOTE. It provides a linear decision boundary and enables coefficient-based interpretation, which is useful for validating that the feature engineering created the expected directional relationships.

Model 2: Random Forest

Random Forest was selected as the ensemble baseline because it handles non-linear relationships, is robust to outliers, and provides built-in feature importance. It is less prone to overfitting than single decision trees and performs well on tabular fraud data. Hyperparameters tuned: n_estimators, max_depth, min_samples_split, max_features.

Model 3: XGBoost (Final Champion)

XGBoost is the industry standard for tabular fraud detection. Its gradient boosting framework sequentially corrects errors from prior iterations, making it particularly effective at detecting the subtle patterns characteristic of sophisticated fraud. Key advantages for this context: native handling of missing values, scale_pos_weight for imbalance, and compatibility with SHAP explainability.

Model 4: Decision Tree (Interpretability Benchmark)

A shallow decision tree (max_depth=6) was trained to generate a human-readable decision ruleset for use by the claims operations team. It is not the champion model but serves as the explainability floor. Any more complex model must demonstrably outperform it to justify the additional complexity.

4.3 Model Performance Comparison

Model	AUC-ROC	PR-AUC	Precision (Fraud)	Recall (Fraud)	F1 (Fraud)	Fi Tim
Logistic Regression	0.8823	0.5841	0.61	0.72	0.66	~8s
Decision Tree (d=6)	0.8241	0.4912	0.55	0.68	0.61	~3s
Random Forest (tuned)	0.9187	0.6893	0.71	0.76	0.73	~4m
XGBoost (tuned) ★	**0.9412**	**0.7318**	**0.76**	**0.79**	**0.77**	~7m

Model Selection Rationale: Why XGBoost

XGBoost outperforms all baselines across every metric. More importantly, it achieves 79% recall on the fraud class at the 0.5 threshold, meaning it correctly identifies 4 out of every 5 fraudulent claims that pass through the pipeline. The 7-minute training time is entirely acceptable for a batch-scored pipeline (claims are scored nightly or at submission, not real-time). Random Forest is close (AUC 0.9187) but XGBoost's PR-AUC advantage of 4.25 percentage points is significant when operating in the high-recall region that matters most for fraud operations.

4.4 XGBoost Hyperparameter Configuration

Parameter	Tuned Value	Search Range	Impact
n_estimators	500	100 – 1000	More trees improve performance to ~500; diminishing returns beyond
max_depth	6	3 – 10	Controls overfitting; depth 6 balances complexity and generalization
learning_rate	0.05	0.01 – 0.3	Lower rate with more estimators gives better generalization
subsample	0.8	0.5 – 1.0	Row sampling per tree, which reduces overfitting
colsample_bytree	0.7	0.5 – 1.0	Feature sampling per tree, which reduces correlation between trees
scale_pos_weight	27.6	Calculated (N_neg/N_pos)	Compensates for class imbalance in addition to SMOTE
reg_alpha (L1)	0.1	0 – 1	Sparse feature weighting that promotes focus on top features
reg_lambda (L2)	1.0	0 – 10	Ridge regularization that prevents coefficient explosion
min_child_weight	5	1 – 20	Minimum sum of instance weight in a leaf, preventing

			overfitting on small fraud clusters
gamma	0.1	0 – 1	Minimum loss reduction required to create a split, promoting conservative tree growth

4.5 Threshold Analysis & Operational Decision Rules

The default 0.5 threshold is not necessarily the right operational threshold. Fraud operations decisions involve an explicit tradeoff between false positives (customer experience damage, unnecessary review cost) and false negatives (undetected fraud, financial loss). I evaluated model performance across thresholds to define the operational zones used in ProTech's claims workflow:

Fraud Score Range	Operational Decision	Precision at Threshold	Recall at Threshold	Business Logic
0.00 – 0.20	Auto-approve (STP)	N/A	N/A	Claims with less than 20% fraud probability are cleared automatically. This tier is expected to cover approximately 72% of all claims, freeing adjuster capacity.
0.20 – 0.60	Standard adjuster review	~62%	~91%	Normal claims process where the adjuster performs standard checks. This tier captures the high-recall zone.
0.60 – 0.85	Enhanced review	~76%	~79%	Senior adjuster or fraud analyst review required. Payment hold until investigation complete.

>0.85	Escalate + payment hold	~89%	~58%	Immediate escalation to fraud team. Claim flagged in partner performance record. Threshold chosen to maximize precision.

This tiered decisioning framework means the model is never a black-box accept/reject system. It is an intelligent routing tool that directs claims to the right review level. Auto-approvals handle ~72% of claim volume (lowering operational cost); the top 5% of high-score claims receive maximum scrutiny (maximizing fraud recovery).

STEP 5

Critical Thinking: Ethical AI & Bias Auditing

5.1 Model Explainability via SHAP Analysis

SHAP (SHapley Additive exPlanations) values were computed for the XGBoost champion model using a random sample of 5,000 test records, stratified across 2,500 fraud and 2,500 legitimate cases. SHAP provides a theoretically grounded attribution of each feature's contribution to the model's output for each individual prediction.

Top 10 Features by Mean |SHAP| Value

Rank	Feature	Mean |SHAP|	Direction	Business Interpretation
1	V126	0.312	High V126 indicates lower fraud risk	Device fingerprint consistency. Claims submitted from unrecognized devices, where V126 is near 0, are high risk. This is consistent with retail-level batch filing.
2	V130	0.287	High V130 indicates lower fraud risk	Session behavioral signal. Legitimate users show consistent session patterns; fraud submissions show anomalous behavior.
3	log_claim_amt	0.241	High amt → High fraud risk	Large claims are disproportionately fraudulent. Every 10× increase in claim amount increases log-odds of fraud by ~0.9.
4	days_since_policy	0.198	Low days → High fraud risk	The single most actionable business signal. Claims in week 1–2 are 4× more likely to be fraudulent than week 8+ claims.
5	V136	0.187	See SHAP plot for directional impact	Behavioral metadata correlated with submission environment authenticity.
6	claim_velocity	0.156	High velocity correlates with higher risk	Partner-level or customer-level repeat claiming. This is our primary organized ring detection feature.
7	email_risk_score	0.143	High score correlates with higher risk	Disposable or anonymous email domains are strongly associated with fraudulent policy activation.
8	card1	0.138	Certain partners carry elevated risk	Individual partner fraud risk. Certain card1 values have consistently high SHAP

				contributions across all their claims.
9	mismatch_score	0.121	High mismatch correlates with higher risk	Identity inconsistency at claim time. Multiple mismatches in customer fields strongly predict fraud.
10	amt_to_device_ratio	0.098	A ratio above 1.0 correlates with higher risk	Over-claiming detector. Claims where payout approaches or exceeds device market value are flagged.

The SHAP summary plot (available in the accompanying Jupyter notebooks) confirms that the model's decisions are driven by economically sensible fraud signals. There are no spurious correlations or counterintuitive patterns in the top features, which is a critical quality gate for production deployment. Importantly, protected attributes (age, gender, race, geographic origin) do not appear in the top 30 SHAP contributors, confirming that the model is not making fraud decisions based on demographic proxies.

5.2 Bias Detection & Fairness Audit

Fraud detection models carry a specific fairness risk: they can systematically over-flag certain demographic groups as suspicious, creating a discriminatory impact even when the protected attribute is not explicitly included. In the ProTech context, the most relevant sensitive attributes are geography (customer region), retail channel (partner), and payment method (which can proxy for socioeconomic status).

Fairness Metrics Evaluated

Fairness Metric	Definition	Result	Pass/Fail	Threshold
Demographic Parity	Difference in positive (fraud) prediction rates between geographic groups (addr1 high-risk vs. low-risk regions)	0.089	PASS	<0.10
Equalized Odds: TPR	Difference in True Positive Rate (fraud recall) between card4 groups (visa vs. prepaid)	0.071	PASS	<0.10
Equalized Odds: FPR	Difference in False Positive Rate between geographic groups	0.034	PASS	<0.10
Disparate Impact Ratio	Ratio of positive prediction rates: low-risk region / high-risk region	0.847	CAUTION	0.80–1.20 is acceptable; 0.847 is within range but warrants monitoring
Predictive Parity	Difference in precision across partner groups, comparing the top	0.052	PASS	<0.10

	20% against the bottom 80% by volume			

Bias Audit Findings

- Geographic fairness (PASS): The model does not disproportionately flag claims from specific provinces. The Disparate Impact Ratio of 0.847 is within the 80% rule threshold, but the audit recommends monitoring this metric quarterly as claim volumes shift by region.
- Partner channel bias (PASS with caveat): Legitimate customers of high-fraud-rate partners may experience higher false positive rates because the partner fraud rate feature contributes to their score. This is operationally correct, since a customer whose retail partner has a 15% fraud rate carries elevated risk through no fault of their own, but it does create a fairness concern. Mitigation: apply partner-level fraud rate as a feature rather than a hard multiplier, and maintain human review for enhanced-review-tier claims rather than auto-rejecting.
- Payment method proxy (MONITORED): Prepaid card holders (card4=prepaid) are flagged at 2.1× the rate of credit card holders, which may disproportionately impact lower-income customers. The model was re-tested with card4 excluded. AUC dropped only 0.009 points, from 0.9412 to 0.9323. Given this minimal performance loss, excluding card4 is recommended for production deployment to eliminate the socioeconomic proxy risk.

5.3 Model Limitations

Limitation	Nature	Impact	Mitigation
Dataset mismatch	The IEEE-CIS dataset covers e-commerce financial fraud rather than device insurance claims. Feature mappings are analogical, not exact.	High	Deploy as a pilot on 3-month holdout data from ProTech's actual claims, then retrain on proprietary data before full production rollout
Class imbalance (3.5% vs. ~8% ProTech actual)	SMOTE and scale_pos_weight are tuned for a 3.5% fraud rate. ProTech's higher actual fraud rate will shift the optimal threshold.	Medium	Recalibrate threshold using ProTech's actual fraud rate after pilot data collection
Feature drift	Fraud patterns evolve. A model trained on 2019 data may not capture emerging fraud tactics such as AI-generated identity documents	Medium	Implement monthly model performance monitoring with drift detection (PSI on top features); retrain annually
Cold start problem	New customers and new retail	Medium	Apply 'new entity' flag; route all new partner claims to standard review for first 90 days

	partners have no behavioral history, so the claim_velocity and partner_fraud_rate features will be zero or missing		
Explainability vs. performance gap	XGBoost is less interpretable than Logistic Regression. Adjuster questions about why a claim was flagged require SHAP explanations rather than raw model internals.	Low	SHAP-based explanation API generates plain-language rationale for every flagged claim
Prepaid card proxy bias	card4=prepaid may proxy for socioeconomic status	Medium	Exclude card4 from the production model. The 0.009 AUC cost is acceptable given the benefit of eliminating discrimination risk

5.4 Mitigation Strategies & Ethical Framework

- Human-in-the-loop design: The model does not automatically reject claims. It routes claims to the appropriate review tier. All enhanced-review and escalated claims are reviewed by a human adjuster before any adverse action is taken.
- Threshold governance: The fraud score threshold for each tier is a business decision, not a model decision. It is reviewed quarterly by a cross-functional committee including claims operations, legal/compliance, and customer experience.
- Model card publication: A model card documenting training data, intended use, known limitations, and fairness metrics is maintained as a living document and shared with regulatory bodies on request.
- Customer appeal right: Any customer whose claim is denied based on model flagging has a defined escalation path to a senior adjuster with full SHAP explanation provided in plain language.
- Adverse action notice: Customers whose claims are declined receive an explanation that does not reference the AI model score. They are informed of the specific inconsistencies identified during review, such as the claim date preceding the policy activation period.

STEP 6

Final Presentation & Communication

Two presentation decks were developed as separate deliverables accompanying this report:

Deck	Audience	Format	Slide Count	Key Focus
Technical Presentation	Data scientists, ML engineers, AIM peers	Jupyter Notebook Slides + PDF export	11 slides	Methodology, model comparison, SHAP analysis, bias audit details, pipeline architecture
Business Presentation	Executive leadership, investors, board	PowerPoint / PDF	10 slides	Fraud problem cost, model ROI, implementation roadmap, risk management, recommended actions

The Technical Presentation follows the ML lifecycle structure: problem → data → EDA → modeling → evaluation → explainability → deployment. It is structured for a technical peer audience who can evaluate methodology choices.

The Business Presentation opens with the financial cost of undetected fraud (USD 192K annual exposure at ProTech's scale), presents the model's business case as an investment decision comparing USD 25K deployment cost against USD 110 to 130K in annual savings, and closes with a phased implementation roadmap. It contains no mathematical notation. All metrics are translated into operational and financial language.

Both decks are available in the GitHub repository under /reports/. The outline for each deck is documented in the Technical Appendix of this report.

STEP 7

GitHub Repository & Reproducibility

7.1 Repository Structure

The complete capstone project is hosted as a public GitHub repository at:

`github.com/ AzzmanV/insurtech-fraud-detection`

Directory / File	Contents	Purpose
README.md	Project overview, setup instructions, results summary, architecture diagram	Project front door that immediately communicates value and how to run the code
requirements.txt	All Python dependencies with pinned versions	Reproducible environment setup
notebooks/01_EDA.ipynb	Data loading, cleaning, missing value analysis, EDA visualizations	Exploratory analysis with narrative commentary
notebooks/02_Feature_Engineering.ipynb	Feature construction, VIF analysis, SHAP-based selection, PCA	Feature engineering pipeline with justification at each step
notebooks/03_Modeling.ipynb	Model training, hyperparameter tuning, CV results, model comparison table	Full modeling pipeline with all models reproducible from seed
notebooks/04_Evaluation_SHAP.ipynb	Test set evaluation, confusion matrices, SHAP plots, fairness metrics	Final evaluation and explainability, submission-ready
src/data_preprocessing.py	Cleaning, imputation, and encoding functions	Modular preprocessing functions reusable for production
src/feature_engineering.py	All engineered feature construction functions	Clean separation of feature logic from model logic
src/model_training.py	Training, CV, and saving functions	Reproducible model training with config-driven runs
src/evaluate.py	Evaluation metrics, threshold analysis, fairness audit	Standalone evaluation module
data/raw/	Pointer file and Kaggle download instructions. Data is not committed due to license restrictions.	Documents data source and version

data/processed/	Processed feature set, post-engineering and pre-modeling	Allows modeling notebook to run without re-running EDA
models/	Saved XGBoost, Random Forest, and Logistic Regression model files in .pkl format, along with the SHAP explainer	Model artifacts for inference and explanation
reports/technical_presentation.pdf	Technical slide deck PDF	Submission deliverable
reports/business_presentation.pdf	Business slide deck PDF	Submission deliverable
reports/model_card.md	Model card: intended use, limitations, fairness metrics	Ethical AI documentation
deployment/app.py	FastAPI inference endpoint	Optional Step 8: production deployment demo
deployment/Dockerfile	Containerized deployment configuration	Reproducible deployment environment

7.2 Reproducibility Standards

- All notebooks use RANDOM_STATE = 42 throughout. Running notebooks in sequence (01 → 02 → 03 → 04) produces identical results.
- SMOTE is applied inside the sklearn Pipeline with the training fold only, confirming no data leakage into the test set.
- Model artifacts are saved with joblib and include the full Pipeline object with imputer, scaler, and model. Inference can be performed on raw features without manual preprocessing.
- requirements.txt pins all library versions. A conda environment YAML (environment.yml) is also provided for conda users.
- A Makefile at repo root provides make train, make evaluate, and make deploy commands for one-command reproducibility.

7.3 Optional Step 8: Deployment via FastAPI

A minimal FastAPI inference endpoint is deployed as a local Docker container. It accepts a JSON payload of claim features and returns a fraud probability score, a risk tier classification (STP / Standard / Enhanced / Escalate), and a plain-language SHAP explanation of the top 3 factors driving the score.

Endpoint	Method	Input	Output
/predict	POST	JSON: claim features (TransactionAmt, TransactionDT, ProductCD, card1, P_emaildomain, engineered features)	JSON: {fraud_probability, risk_tier, top_3_shap_factors, recommended_action}
/health	GET	None	Model version, last trained date, current AUC-ROC from validation

/explain/{claim_id}	GET	Claim ID (for stored predictions)	Full SHAP waterfall explanation for specific claim

A screen recording demo of the FastAPI endpoint is included in the repository under /deployment/demo.gif.

7.4 Optional Step 9: Generative AI Integration

A LLM-powered claims triage assistant was prototyped using the Anthropic Claude API. The assistant receives the model's SHAP explanation (top fraud factors + their directional impact) and generates a plain-language fraud alert narrative for the claims adjuster. For example:

> **GenAI-Generated Fraud Alert Example**
>
> FRAUD ALERT: Claim XC-20847, Risk Score: 0.91 (ESCALATE) This claim shows three converging risk signals that warrant immediate investigation: 1. TIMING: The claim was filed 6 days after policy activation. Historical data shows that claims within the first 14 days have a 4× higher fraud probability than those filed after 60 days. 2. DEVICE MISMATCH: The claim amount ($823) exceeds the average replacement cost for the ProductCD category (W = Water Damage, avg. $600), suggesting possible over-claiming or misclassification. 3. PARTNER RISK: This claim comes from Retail Partner #4419, which has a rolling 30-day fraud rate of 14.2%, significantly above the network average of 3.5%. Recommended action: Hold payment and escalate to fraud analyst for partner audit. Do not decline. Obtain additional documentation (purchase receipt, device serial verification) before making a coverage decision.

The GenAI component is documented in /notebooks/05_GenAI_Triage_Demo.ipynb. It converts model outputs into adjuster-ready fraud alerts, a practical application that bridges ML output and operational workflow.

BUSINESS IMPACT

ROI Analysis & Business Value

Financial Impact Model

The following ROI model is based on ProTech Devices Asia's estimated portfolio parameters and the model's validated performance metrics. Figures are estimates and will be refined after the pilot deployment.

Parameter	Assumption	Source
Annual claims volume	5,400 claims/year (~450/month)	ProTech internal claims data (estimated)
Average claim payout	$440 per legitimate claim	Industry benchmark, device protection
Annual fraud rate (estimated)	8% of claims	Industry range: 6–12%; conservative assumption
Annual fraudulent claims	432 claims/year	5,400 × 8%
Annual fraud exposure (gross)	$193,600	432 × $448 avg. fraud claim (higher than legit avg.)
Model fraud recall (at 0.5 threshold)	79%	Validated on test set
Model precision (at 0.5 threshold)	76%	Validated on test set
Fraud claims caught by model	341 of 432	432 × 79%
Fraudulent payouts prevented	~$152,700/year	341 × $448
False positives (legitimate claims flagged)	~108 additional reviews/year	At precision 76%: for every 341 true positives, ~108 false positives
Cost of false positive (re-review)	$25 per claim (adjuster time)	ProTech operational cost estimate
False positive cost per year	$2,700	108 × $25
Net annual fraud savings	~$150,000	$152,700 − $2,700
Model development cost (one-time)	$35,000	Includes data engineering, modeling, deployment, testing
Annual maintenance cost	$8,000/year	Model monitoring, quarterly retraining, support
Year 1 ROI	304%	($150K − $8K operational − $35K development) / $43K total cost
Year 2 ROI (ongoing)	1,600%+	($150K − $8K) / $8K ongoing cost

Conservative ROI Note

This model assumes only direct fraud prevention savings. It excludes: (1) operational efficiency gains from auto-approving 72% of claims via straight-through processing (STP), reducing adjuster workload by an estimated 30%; (2) reduced claim processing time improving customer NPS; (3) retail partner audit savings from the partner_fraud_rate signal enabling proactive channel management. Including these, total first-year business value likely exceeds $200,000 against a $43,000 investment.

Implementation Roadmap

Phase	Timeline	Activities	Success Criteria
Phase 1: Pilot	Month 1–3	Deploy the model in shadow mode, scoring claims without acting on the scores. Collect labeled outcomes. Compare model scores to adjuster decisions.	Model AUC >0.85 on ProTech's own data. Adjuster agreement rate >70% on high-score claims.
Phase 2: Threshold Calibration	Month 4	Set operational thresholds based on ProTech's actual fraud rate. Calibrate STP zone to minimize false positive rate.	False positive rate <5% of total legitimate claims. STP zone covers >65% of claim volume.
Phase 3: Soft Launch	Month 5–6	Enable routing rules for STP and Enhanced Review tiers. The Escalate tier still requires manual approval.	10% reduction in average claim processing time. Zero customer complaints attributable to false positive adverse actions.
Phase 4: Full Production	Month 7+	Full operational deployment including the partner fraud monitoring dashboard and monthly model performance reviews.	15% reduction in fraudulent payouts vs. pre-model baseline. Quarterly model retraining cycle established.

CONCLUSION

Conclusion

When I started building this project, my goal was not to produce a good machine learning model. My goal was to answer a question I face operationally: how do we stop paying for fraud that we could have detected?

The answer, it turns out, is both more attainable and more nuanced than I expected. The XGBoost model achieves an AUC-ROC of 0.9412, a performance level that would have felt ambitious at the start of this program. But the more valuable output is not the model itself. It is what the model reveals: that device insurance fraud in Asia is not random. It follows patterns. Claims filed in week one, by customers with unverifiable email addresses, submitted from retail partners with histories of elevated claim rates, for amounts that approach device replacement value. These claims are telling a story before a human adjuster ever opens the file.

The SHAP analysis gives us the vocabulary to articulate that story. The fairness audit ensures we are not telling it at the expense of legitimate customers who happen to share characteristics with fraudsters. The tiered decisioning framework ensures that the model is a tool that improves human judgment, not a system that bypasses it.

I am not treating this project as an academic exercise. The pipeline developed here, adapted to ProTech's proprietary claims data, is the foundation of what I intend to build in the second half of 2025. The ROI case is clear. The implementation roadmap is realistic. The ethical safeguards are in place.

The question is no longer whether AI can help detect insurance fraud in device protection. It demonstrably can. The question is how quickly we move from proof of concept to production, and whether we do it carefully enough to earn the trust of the customers, regulators, and partners who depend on us to get it right.

Mark Eliezer M. Villola
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AIM Postgraduate Diploma in AI & Machine Learning | Pillar 5 Capstone

APPENDIX A

Dataset Identification & Selection Analysis

A.1 Candidate Dataset Evaluation

Three public datasets were evaluated for their fitness as the primary data source for this InsurTech fraud detection capstone. The evaluation criteria were: (1) structural relevance to insurance claims, (2) feature richness for fraud signal engineering, (3) class imbalance characteristics reflective of real fraud environments, and (4) data quality and documentation.

Candidate 1: IEEE-CIS Fraud Detection (SELECTED)

- Source: Kaggle (Vesta Corporation / IEEE Computational Intelligence Society, 2019)
- Size: 590,540 transactions × 394 features (transaction + identity tables)
- Fraud rate: 3.5%, which is a realistic imbalance for supervised fraud detection
- Key strengths: Rich behavioral V-features (device fingerprinting, session signals), match indicator features (M1–M9), email domain fields, and partner/channel identifiers. All are directly mappable to the insurance claims context
- Adaptation required: Feature re-interpretation from e-commerce to insurance claim context; custom engineered features to create InsurTech-specific signals
- Conclusion: Best available public proxy for device insurance fraud and therefore selected as the primary dataset

Candidate 2: PaySim (Rejected)

- Source: Kaggle (synthetic simulation of mobile money transactions)
- Size: 6.3M rows × 11 features
- Fraud rate: 0.13%, which is extremely low and requires aggressive resampling
- Key weakness: Synthetic data with only 11 features; no behavioral signals; fraud patterns are artificially simple. Only transfer and cash-out transactions are fraudulent, meaning a simple rule detects 100% of fraud. This dataset would not demonstrate ML modeling skills.
- Conclusion: Rejected. Too simple, synthetic, and lacking sufficient feature richness

Candidate 3: Credit Card Fraud Detection from ULB (Rejected)

- Source: Kaggle (Worldline and ULB Machine Learning Group)
- Size: 284,807 transactions × 30 features (28 PCA-transformed, 2 original)
- Fraud rate: 0.17%
- Key weakness: All behavioral features are PCA-transformed and anonymized, making it impossible to map them to InsurTech domain concepts or create domain-specific engineered features. The dataset is excellent for benchmarking algorithm performance but provides no opportunity to demonstrate domain knowledge or feature engineering creativity.
- Conclusion: Rejected. Anonymization prevents InsurTech adaptation and the dataset is insufficient for domain-specific feature engineering

A.2 Adaptation Logic: IEEE-CIS → Device Insurance Claims

The core insight enabling this adaptation is that both e-commerce payment fraud and device insurance claim fraud share fundamental behavioral structures: a fraudster must establish a false identity, create a plausible transaction/claim event, and extract value before detection. The specific mechanics differ, but the detectable signals, including timing anomalies, channel behavior, identity inconsistency, and amount patterns, are structurally similar. This makes the IEEE-CIS dataset a valid proxy while the adaptation logic is documented transparently as a project assumption.

APPENDIX B

Presentation Slide Outlines

B.1 Technical Presentation Slide Outline

Slide	Title	Key Content
1	Title & Context	Project title, author, ProTech context, AIM affiliation
2	The Problem in Data Terms	Fraud rate, class imbalance visualization, financial exposure calculation
3	Dataset & Adaptation Logic	IEEE-CIS selection rationale, InsurTech feature mapping table
4	EDA Insights	Key findings: claim amount distribution, time-to-claim analysis, V-feature KDE separation
5	Feature Engineering	10 engineered features table, VIF results, feature selection process
6	PCA & t-SNE Visualization	Explained variance plot, t-SNE fraud cluster visualization
7	Model Comparison	4-model comparison table: AUC-ROC, PR-AUC, Precision, Recall, F1
8	XGBoost Tuning and Configuration	Hyperparameter search results, learning curves, final configuration
9	SHAP Analysis	SHAP summary plot, top 10 feature SHAP values, individual claim SHAP waterfall
10	Bias & Fairness Audit	Fairness metrics table, geographic parity results, card4 proxy risk discussion
11	Deployment Architecture & Next Steps	Claims workflow integration diagram, FastAPI endpoint, monitoring plan, retraining cadence

B.2 Business Presentation Slide Outline

Slide	Title	Key Content
1	The Fraud Problem: What It Costs Us	USD 192K annual exposure, 3 fraud patterns, operational gap in detection
2	Why Traditional Review Fails	Manual review limitations: adjuster capacity, threshold rules miss sophisticated fraud, no systematic scoring
3	Our AI Fraud Detection Engine	One-sentence model description, claim scoring flow diagram, 4-tier decisioning
4	The Data: What the Model Learned From	590K claims-equivalent transactions, 6 months of behavioral data, 10 engineered InsurTech features
5	What the Model Found	Top 3 fraud signals in plain language: timing, claim amount, partner risk score
6	How It Works in Operations	Claims intake → score → route diagram. STP (72% auto-approve), Standard (23%), Enhanced (4%), Escalate (1%)

7	The Results in Numbers That Matter	79% of fraud caught, AUC 0.94, <5% false positive rate, estimated USD 150K annual savings
8	Is It Fair?	Fairness audit summary: passes demographic parity, equalized odds. Human override always available. No auto-rejections.
9	ROI & Investment Case	USD 35K deployment cost vs. USD 150K Year 1 savings, yielding 304% Year 1 ROI and USD 142K Year 2 net benefit
10	Implementation Roadmap & Ask	4-phase roadmap (pilot → calibration → soft launch → production), resource ask, decision required